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PROBLEM STATEMENT #35 | Retail, E-Commerce & Finance

Customizable Subscription Box Scheduler

- 1. Problem Context & Overview** A subscription box portal where subscribers customize monthly product selections (e.g., books, coffee, snacks) based on preference tags, with support for pausing or skipping billing cycles. Target Stakeholders / Actors: Subscriber, Fulfillment Lead

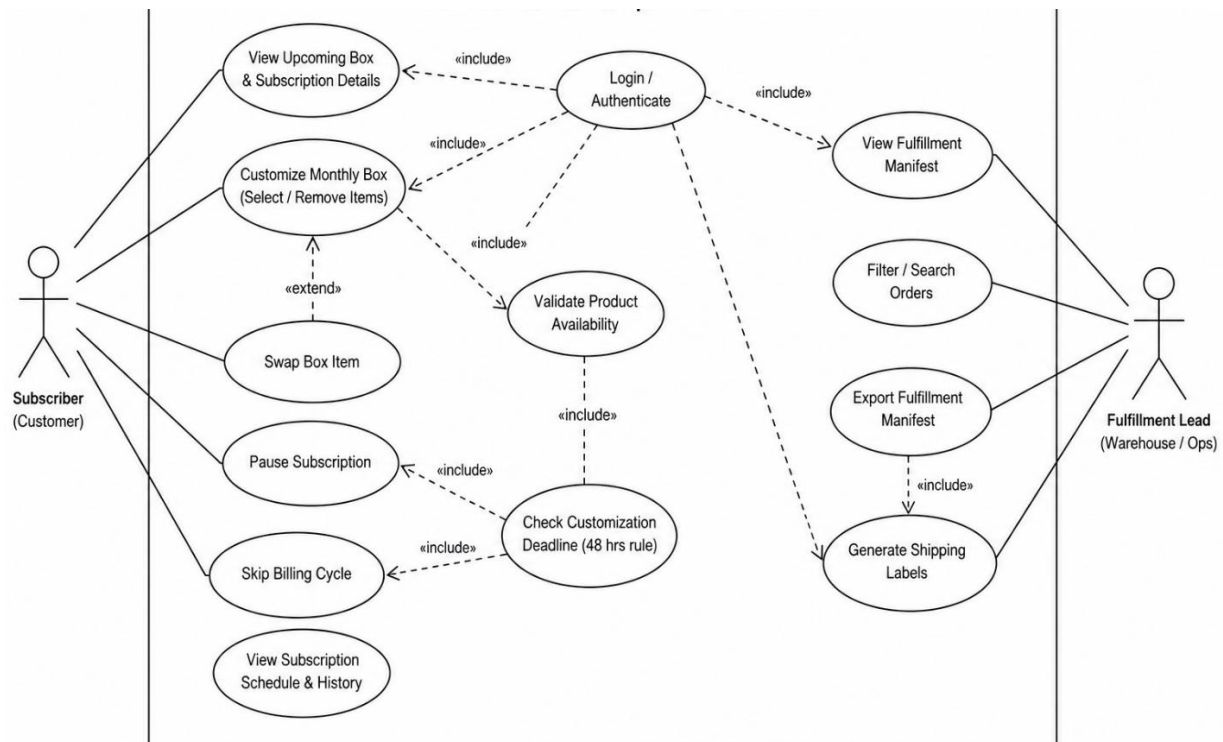
ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	functional	The system shall allow subscribers to customize their upcoming monthly subscription box by selecting, removing, or swapping available products based on their preference tags	High	Pass: Subscriber can select or swap valid products and the updated box is saved. Fail: Invalid or unavailable products are accepted.	Allows subscribers to personalize their monthly box according to their preferences.
FR-002	Functional	The system shall allow subscribers to pause their subscription or skip an upcoming billing cycle up to 48 hours before the monthly billing renewal date.	High	Pass: Pause/skip request made before the 48-hour deadline is accepted and the billing/delivery schedule is updated. Fail: A request after the deadline is incorrectly accepted.	Gives subscribers flexibility while ensuring sufficient time for fulfillment operations
FR-003	Functional	The system shall display the subscriber's upcoming box, selected products, billing renewal date, delivery date, and customization deadline.	Medium	Pass: Correct box details and dates are displayed. Fail: Required subscription or scheduling information is missing or incorrect	Helps subscribers understand their upcoming order and the deadline for making changes.
FR-004	Functional	The system shall automatically refresh the fulfillment manifest whenever a subscriber successfully customizes, pauses, or skips an upcoming box.	High	Pass: The fulfillment manifest reflects the latest product selections and delivery status. Fail: The manifest contains outdated subscription information	Ensures the fulfillment team always works with accurate order information.

FR-005	functional	he system shall allow the Fulfillment Lead to view, filter, and export the monthly fulfillment manifest containing subscriber orders, selected products, and delivery status.	high	Pass: Fulfillment Lead can view, filter, and export accurate order information. Fail: Orders are missing, duplicated, or contain incorrect product selections.	Helps the fulfillment team efficiently prepare and ship the correct subscription boxes.
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NON FUNCTIONAL REQUIREMENTS

id	type	description	priority	Acceptance criteria	Rationale
NAR-001	Performance and scalability	The system shall generate and export the fulfillment manifest and shipping labels for 10,000 monthly subscription boxes in under 1 minute under peak load.	High	Pass: Export for 10,000 boxes completes within 60 seconds. Fail: Export takes more than 60 seconds.	Ensures the system can efficiently handle a large number of subscribers during monthly fulfillment.
NAR-002	Security and reliability	The system shall provide authenticated and role-based access to subscriber and fulfillment data and ensure subscription updates are processed reliably without data loss or corruption.	high	Pass: Subscribers access only their own data and Fulfillment Leads access authorized data. Fail: Unauthorized access or inconsistent data occurs.	Protects customer information and prevents incorrect billing, customization, or fulfillment operations.

UML Use-Case Diagram: >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.



Use-Case Flow Specification: >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario >> One Alternate Flow.

USE-CASE FLOW SPECIFICATION

Use Case: Customize Monthly Subscription Box

Use Case ID: UC-01

Primary Actor: Subscriber

Goal: To allow the subscriber to customize the products in their upcoming monthly subscription box before the customization deadline.

Trigger: The subscriber selects the “Customize Monthly Box” option.

Main flow:

1. Preconditions

1. The subscriber has an active subscription.
2. The subscriber is logged into the system.
3. An upcoming monthly subscription box is available.
4. The customization deadline has not passed.
5. The upcoming billing cycle has not yet been finalized for fulfillment.

2. Postconditions

Success:

1. The subscriber's selected products are updated successfully.
2. The updated box configuration is saved in the system.
3. The fulfillment manifest is refreshed with the updated selection.
4. The Fulfillment Lead can view the updated order.
5. The subscriber receives a confirmation message.

Failure:

1. The original box selection remains unchanged.
2. The fulfillment manifest is not incorrectly modified.
3. The subscriber receives an appropriate error message.

3. Main Success Scenario

1. The Subscriber logs into the subscription portal.
2. The system displays the subscriber's upcoming box and subscription details.
3. The Subscriber selects "Customize Monthly Box".
4. The system checks whether the customization deadline is valid.
5. The system displays available products based on the subscriber's preference tags.
6. The Subscriber selects, removes, or swaps products in the upcoming box.
7. The system checks the availability of the selected products.
8. The system displays the updated box details for confirmation.
9. The Subscriber confirms the customization.
10. The system saves the updated product selection.
11. The system refreshes the fulfillment manifest with the updated box information.
12. The system displays a confirmation message to the Subscriber.
13. The Fulfillment Lead can view the updated order in the fulfillment manifest.

4. Alternate Flow**AF-01: Customization Deadline Has Passed**

1. At Step 4, if the customization deadline has passed, the system prevents the Subscriber from modifying the upcoming box.

2. The system displays the message: “Customization is no longer available for this billing cycle. Changes must be made at least 48 hours before the billing renewal date.”
3. The existing box selection remains unchanged.
4. The fulfillment manifest is not modified.
5. The use case ends.

5. Summary

The use case allows a Subscriber to customize an upcoming monthly subscription box by selecting, removing, or swapping products. The system validates the customization deadline and product availability before saving the changes. Once the customization is confirmed, the fulfillment manifest is automatically updated so that the Fulfillment Lead has the latest and correct order information.